

Support Vector Machine (SVM)

→ It is a supervised machine learning algorithm used for Classification

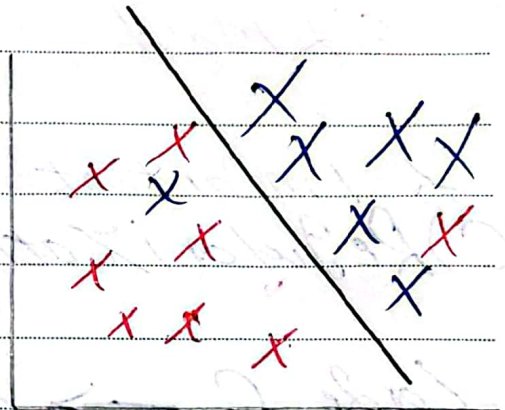
→ Core Idea: SVM tries to find the best decision boundary that separates class by class

Maximizing the distance between the boundary and the closest data points from each class

Types of SVM

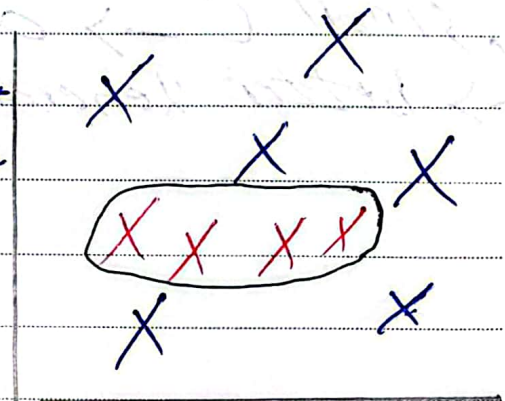
[1] Linear SVM

used when data is linearly separable



[2] Non-Linear SVM

used when data cannot be separated by a straight line.



* Kernel Trick

→ The Kernel Function transforms data into a higher-dimensional space where it becomes linearly separable

Common Kernels:

Linear

Polynomial

RBF

Sigmoid

* Important Hyper Parameters:

(C) Regularization Parameter:

Large C → less Tolerance for errors
 ↳ risk for overfitting

small C → allows more errors
 ↳ better generalization